# MIRROR: A Hierarchical Benchmark for Metacognitive Calibration in Large Language Models

Anonymous Authors
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**Abstract**

We introduce MIRROR, a benchmark comprising eight experiments across four metacognitive levels that evaluates whether large language models can use self-knowledge to make better decisions. We evaluate 16 models from 8 labs across approximately 250,000 evaluation instances using five independent behavioral measurement channels. Core experiments are run across the full model roster; experiments with specialized infrastructure requirements report explicitly marked model subsets. We find two phenomena with direct implications for agentic deployment: (1) compositional self-prediction fails universally—the Compositional Calibration Error ranges from 0.500 to 0.943 on the original 15-model Exp3-v1 set (and 0.434 to 0.758 on the balanced 16-model Exp3-v2 expansion), indicating that models cannot predict their own performance on multi-domain tasks, and (2) models exhibit above-chance but imperfect domain-specific self-knowledge yet systematically fail to translate even this partial awareness into appropriate agentic action-selection—external metacognitive control reduces the Confident Failure Rate from 0.600 to 0.143 (76% reduction at temperature 0; mean 70% at temperature 0.7 across 5 models from 4 labs). Providing models with their own calibration scores produces no significant improvement ($p > 0.05$); only architectural constraint is effective. This suggests that external metacognitive scaffolding—not improved self-knowledge—is the path to safer autonomous AI systems. MIRROR and all data are publicly available.

## 1 Introduction

The deployment of large language models (LLMs) as autonomous agents presupposes a form of metacognitive competence: the model must not only know *what* it knows, but must translate that awareness into appropriate action—proceeding when confident, escalating to tools or human review when uncertain. This assumption underlies agentic architectures built on self-reflection loops, confidence-based routing, and Constitutional AI self-monitoring [Perez et al., 2022]. It is foundational to the enterprise of agentic AI safety. It has not been systematically tested across multiple metacognitive levels.

Prior work has studied LLM calibration [Kadavath et al., 2022, Kapoor et al., 2024], uncertainty estimation [Lin et al., 2024], and situational awareness [Laine et al., 2024]. But these address different questions: calibration measures whether confidence matches accuracy; situational awareness measures whether models know *what they are*. While recent work has begun exploring whether models can use self-knowledge for better decisions [Schmied et al., 2025, Qiao et al., 2025], no benchmark has systematically tested the full monitoring-to-control pipeline across multiple metacognitive levels—whether the bridge from metacognitive *monitoring* to metacognitive *control* [Nelson and Narens, 1990] is intact.

We introduce MIRROR, a hierarchical benchmark spanning four metacognitive levels (atomic self-knowledge, cross-domain transfer, compositional prediction, and adaptive self-regulation), eight experiments, and five independent behavioral measurement channels. We evaluate 16 models from 8

**Table 1:** Comparison of MIRROR with prior benchmarks for LLM self-knowledge and metacognition.

| Feature | SAD (Laine 2024) | Barkan et al. (2025)‡ | Kadavath et al. (2022) | Steyvers et al. (2025) | MIRROR (Ours) |
|---|---|---|---|---|---|
| Venue | NeurIPS D&B '24 | NeurIPS WS '25 | arXiv | arXiv | NeurIPS E&D '26 |
| Focus | Situational awareness | Self-capability prediction | Calibration (P(IK)) | Metacognition + uncertainty | Metacognitive cal. → agentic action |
| Models | 16 | ∼15 | ∼10 | 2 families | 16 |
| Task categories | 7 | 3 | — | 6 datasets | 8 experiments |
| Total items | 13K questions | ∼1K tasks | — | 12K+ items | 5,597 tasks† |
| Metacognitive levels | 1 (awareness) | 1 (confidence) | 1 (calibration) | 1 (calibration) | 4 (L0–L3) |
| Behavioral channels | 1 (MC/free) | 1 (verbal) | 1 (verbal) | 1 (verbal) | 5 (wager, opt-out, tool, diff., natural) |
| Agentic connection | No | Yes (SWE-bench) | No | No | Yes (4 cond., 3 par.) |
| Headline finding | Partial sit. awareness | LLMs over-confident | *Mostly* calibrated | Task-specific metacognition | Knowing-doing gap (62% CFR reduction) |
| Public dataset | Yes | No | No | No | Yes |

†5,000 domain questions + 597 agentic tasks. Evaluated across 16 models, 5 channels, and multiple conditions, yielding ∼250K total evaluation instances.

‡Workshop paper; included for comparison as the most directly relevant prior work on LLM self-capability prediction.

labs across approximately 250,000 evaluation instances.[1]

Three findings emerge. First, compositional self-prediction fails universally: the Compositional Calibration Error (CCE) ranges from 0.500 to 0.943 on the original 15-model Exp3-v1 set and 0.434 to 0.758 on the balanced 16-model Exp3-v2 expansion—models that calibrate accurately in individual domains completely fail to compose those calibrations for multi-domain tasks. Second, the knowing-doing gap: models exhibit partial but above-chance self-knowledge—accurately identifying relative domain strengths and weaknesses—yet do not act on it. External constraint reduces the Confident Failure Rate (CFR) from 0.600 to 0.143 (76% reduction); providing models with their own calibration scores produces no significant improvement. Third, self-knowledge is domain-atomic: the Transfer Influence Index ranges from 0.019 to 0.175 across 11 models, indicating that calibration does not meaningfully cross domain boundaries.

**Contributions.**

- The MIRROR benchmark: a reusable evaluation framework comprising 8 experiments across 4 metacognitive levels, 5 behavioral channels, 5,000 domain-stratified questions, and 597 composite agentic tasks. The full benchmark runs on a new model in ∼3 hours via API; all prompts, scoring code, and analysis pipelines are publicly released.
- Two empirical findings—universal compositional failure (CCE 0.500–0.943) and the knowing-doing gap (76% CFR reduction at $t=0$; mean 70% at $t=0.7$ across 5 models from 4 labs)—quantifying the cost: ∼562 of 1,000 agentic tasks fail silently without external constraint vs. ∼214 with constraint.
- Evidence that external metacognitive scaffolding, not better self-knowledge, is the path to safer agentic systems.

Table 1 positions MIRROR relative to prior work.

## 2 Related Work

**LLM Calibration.** Kadavath et al. [2022] showed that LLMs can predict their own answer correctness above chance on knowledge-intensive tasks. Kapoor et al. [2024] demonstrated that fine-tuning is needed for reliable calibration. Steyvers et al. [2025] found that metacognitive ability is task-specific, not general—a finding consistent with MIRROR's domain-level analysis. Wang et al. [2025]

[1]The 250,000 figure counts evaluation instances (model × channel × condition × question) after quality-control filtering and experiment-specific inclusion criteria. Per-experiment model counts are reported explicitly in Section 4 and Table 4. The underlying unique task content comprises 5,000 domain-stratified questions and 597 agentic tasks.

introduced a framework for quantifying metacognitive ability independently of cognitive ability, using a decoupled evaluation that controls for task difficulty. MIRROR differs from all prior calibration work in testing whether calibration *transfers to action-selection*: we measure the gap between knowing and doing, not the accuracy of knowing alone.

**LLM Self-Knowledge and Situational Awareness.** SAD [Laine et al., 2024] measures whether LLMs know what they are—identity, lifecycle stage, deployment context. Barkan et al. [2025] tested capability prediction on SWE-bench tasks, finding systematic overconfidence. Xiong et al. [2024] found that LLMs are often overconfident when expressing numerical probability estimates. MIRROR is complementary: SAD diagnoses awareness; MIRROR diagnoses the gap between awareness and behavior.

**Agentic Evaluation.** AgentBench [Liu et al., 2024], OSWorld [Xie et al., 2024], and GAIA [Mialon et al., 2023b] measure task success rate in agentic settings. These benchmarks evaluate execution quality conditional on the decision to act. Concurrent work on agentic confidence calibration [Anonymous, 2026a] measures calibration during multi-step trajectories using trajectory-level features. MIRROR differs in measuring metacognitive calibration *before* deployment and testing whether pre-deployment assessment predicts agentic outcomes. More broadly, MIRROR inverts the standard framing: we measure the *decision* itself, treating task execution quality as a downstream variable and calibration-informed decision-making as the primary outcome. Schmied et al. [2025] independently identify a "knowing-doing gap" in LLM decision-making and show that RL fine-tuning can partially narrow it. Qiao et al. [2025] study agentic knowledgeable self-awareness, testing when LLM agents recognize their own knowledge boundaries and act accordingly in tool-use settings. MIRROR extends both by systematically measuring the gap between boundary recognition and appropriate action across 8 domains and 4 escalation conditions. Self-correction approaches—including Reflexion [Shinn et al., 2023], SelfCheckGPT [Manakul et al., 2023], and chain-of-verification [Dhuliawala et al., 2023]—test whether iterative self-reflection improves task performance. MIRROR's Experiment 9 tests a complementary question: whether pre-measured metacognitive profiles predict when self-correction will fail. Selective classification methods [Geifman and El-Yaniv, 2017] provide formal frameworks for deciding when to defer; MIRROR's escalation curve evaluates domain-informed selective classification in agentic settings. Recent work on explicit safety scaffolding [Anonymous, 2026c] demonstrates that externalized check-and-refuse loops outperform implicit reasoning for safety-critical decisions, aligning with MIRROR's finding that architectural constraint outperforms self-knowledge exposure.

**Metacognition in AI.** Metacognitive prompting [Wang and Zhao, 2024] uses metacognitive strategies to improve LLM performance. Augmented language model surveys [Mialon et al., 2023a] document the tool-use landscape but do not address when tool use *should* occur. Concurrent work on metacognition in large reasoning models [Anonymous, 2026b] tests difficulty awareness, confidence adjustment, and strategy revision across reasoning models. MIRROR extends this by connecting metacognitive assessment to agentic deployment through the Experiment 9 escalation design, providing the first benchmark-scale empirical test linking metacognitive assessment to agentic deployment outcomes.

## 3 The MIRROR Benchmark

MIRROR decomposes LLM metacognition into four levels of increasing cognitive complexity, inspired by the hierarchical structure of human metacognitive monitoring and control [Flavell, 1979, Nelson and Narens, 1990]. At each level, we measure behavior through five independent channels to prevent format-specific gaming.

### 3.1 Metacognitive Levels

**Level 0—Atomic Self-Knowledge (Calibration).** Can the model accurately estimate $P(\text{correct})$ for a given question in a given domain? This is the best-studied level [Kadavath et al., 2022, Kapoor et al., 2024] and serves as MIRROR's foundation. Experiment 1 establishes per-model, per-domain calibration profiles across 8 cognitive domains.

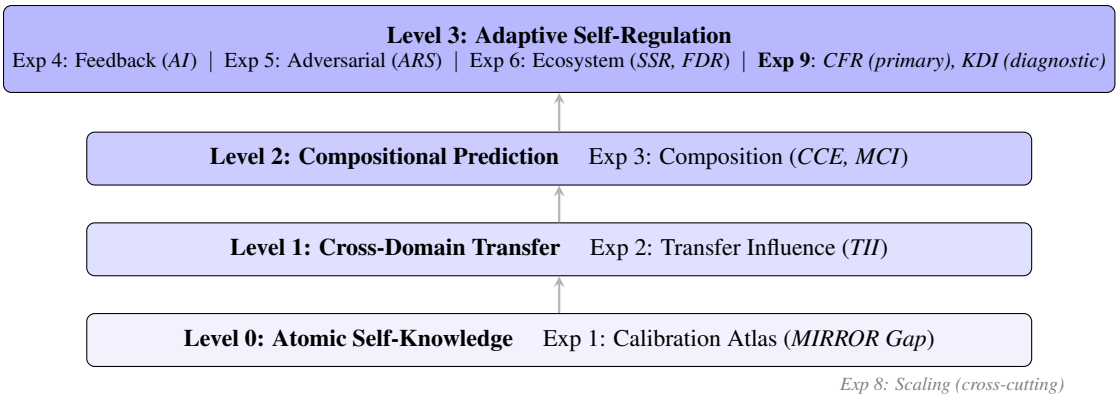


**Figure 1:** The MIRROR experimental hierarchy. Each level builds on measurements from the level below. Experiment 1 produces per-model calibration profiles consumed by all subsequent experiments. The capstone (Experiment 9) tests whether self-knowledge translates to appropriate agentic decisions.

**Level 1—Cross-Domain Transfer.** Does calibration in domain A transfer to tasks implicitly requiring that skill?

**Level 2—Compositional Prediction.** Can the model predict performance on multi-skill tasks?

**Level 3—Adaptive Self-Regulation.** Does the model translate self-knowledge into appropriate action? This is the critical gap between metacognitive *monitoring* (knowing your limits) and metacognitive *control* (acting on that knowledge)—a distinction well-established in human cognition [Metcalfe and Finn, 2008]. Experiments 4, 5, 6, and 9 test whether self-knowledge drives behavior change, feedback adaptation, ecosystem interactions, and agentic decision-making.

### 3.2 Behavioral Measurement Channels

A single confidence elicitation format (*e.g.*, verbal "rate your confidence 1–10") is vulnerable to prompt-specific behavior. MIRROR measures metacognitive behavior through five independent channels, each requiring a separate API call with a fresh context:

1. **Wagering**—the model stakes points proportional to its confidence, with scoring that penalizes overconfident wrong answers.
2. **Opt-out**—the model may decline to answer. Skip rate should correlate with error rate.
3. **Difficulty selection**—given two questions (easy and hard), the model chooses which to attempt.
4. **Tool delegation**—the model may invoke a calculator, search engine, or flag for human review rather than answering directly.
5. **Natural language signals**—hedging language, uncertainty markers, and response length, extracted from unconstrained free-form responses.

The **Metacognitive Convergence Index (MCI)** measures agreement across channels: do all five channels agree about which questions are hard for this model? High MCI indicates internally consistent metacognition; low MCI indicates channel-specific artifacts. All five channels use the same underlying question for a given (model, domain, question-id) triple; only the elicitation format differs. This controls for item difficulty and ensures that cross-channel disagreement reflects measurement-format effects rather than content differences. Table 2 provides a quick reference for all 12 metrics; formal definitions are in Appendix F.

### 3.3 Experiments

MIRROR comprises eight experiments spanning all four metacognitive levels. Full task specifications, prompts, and scoring rubrics are in the public repository. **Experiment numbering note (prominent).** Experiment 7 (mechanistic probing via hidden-state analysis) was pre-registered but omitted from

**Table 2:** MIRROR metric reference. Each metric is formally defined in Appendix F.

| Metric | Exp | Level | Measures |
|---|---|---|---|
| MIRROR gap | 1 | L0 | Wagering vs. natural accuracy divergence |
| MCI | 1,3 | L0/L2 | Cross-channel agreement |
| ECE | 1 | L0 | Within-channel calibration error |
| TII | 2 | L1 | Cross-domain transfer strength |
| CCE | 3 | L2 | Composite prediction error |
| AI | 4 | L3 | Feedback-driven behavior change |
| SAR | 4 | L3 | Sycophancy vs. genuine adaptation |
| ARS | 5 | L3 | Robustness to adversarial framing |
| FDR | 6 | L3 | Flawed premise detection rate |
| SSR | 6 | L3 | Sycophancy separation ratio |
| CFR | 9 | L3+ | Confident failure rate (agentic) |
| KDI | 9 | L3+ | Knowing-doing diagnostic summary (secondary) |

this release due to GPU access constraints; we retain the original numbering for consistency with the pre-registration.

**Experiment 1: Self-Knowledge Atlas (Level 0).** 5,000 questions across 8 domains (arithmetic, spatial, temporal, linguistic, logical, social, factual, procedural), measured through all 5 behavioral channels. 16 models evaluated. Primary metrics: MIRROR gap, MCI.

**Experiment 2: Cross-Domain Transfer (Level 1).** $\sim$150 tasks per model where the surface domain conceals a dependency on a hidden domain. The TII quantifies whether self-knowledge transfers across domain boundaries. 11 models; TII range: 0.019–0.175 (weak-to-minimal transfer).

**Experiment 3: Compositional Self-Prediction (Level 2).** $\sim$25 multi-domain tasks per model in the original v1 set (15 models), plus a balanced v2 expansion (112 tasks over all 28 domain pairs) with 16-model coverage. The CCE measures predicted vs. actual composite accuracy. Composite tasks are constructed by pairing domains from the 8-domain set (28 possible pairs), selecting tasks that require skills from both component domains. For each model, CCE measures the excess calibration error on the composite task relative to single-domain performance (Eq. 5). Primary metric: CCE (see Section 5.2 for results and bootstrap CIs). MCI is also reported with explicit stability diagnostics (Section 6).

**Experiment 4: Adaptation Crucible (Level 3).** 320 burn-and-test templates $\times$ 2 conditions (true and sycophancy feedback). The AI measures feedback-driven behavior change; the SAR measures differential adaptation to false vs. true feedback. 15 models; AI $\approx$ 0 (feedback immunity); SAR range: $-1.52$ to 2.97.

**Experiment 5: Adversarial Robustness (Level 3).** 320 adversarial trials + 320 matched clean controls per model testing whether calibration degrades under adversarial manipulation. 14 models; ARS = 0.937–0.991 (robust).

**Experiment 6: Ecosystem Effect (Level 3).** (a) Sycophancy in multi-agent trust scenarios (SSR). (b) Flawed premise detection (FDR: 0.000–1.000). (c) Value robustness under incremental pressure. 16 models evaluated.

**Experiment 8: Scaling Analysis (Levels 0–3).** Within-family comparison across Llama 3.1 8B $\rightarrow$ 70B $\rightarrow$ 405B and a generation comparison (Llama 3.1-70B vs. 3.3-70B). Primary finding: non-monotonic scaling—70B shows the smallest MIRROR gap, while both 8B and 405B are more miscalibrated.

**Experiment 9: The Knowing-Doing Gap (Level 3+).** 597 agentic tasks (297 fixed + 300 tailored) $\times$ 4 conditions $\times$ 3 paradigms, evaluated on all 16 models from 8 labs. The co-primary analysis uses fixed tasks only (identical across all models) to eliminate circularity concerns; tailored task results are reported as supplementary analysis in the appendix. Each task has two components requiring different domain skills. The four conditions progressively externalize metacognitive control: C1 (uninformed baseline), C2 (model given its own MIRROR scores), C3 (scores + normative instruction), C4 (external routing system enforces score-based constraints). Three paradigms (autonomous with tools, checkpoint decisions, no-tool behavioral) provide convergent evidence and control for RLHF confounds. Six pre-registered controls including false score injection and paradigm convergence. Weak and strong domains are identified from Experiment 1 calibration profiles using a pre-specified `median_or_bottom_k` policy: a domain is classified as "weak" for model $m$ if Nat.Acc$(m, d)$ falls below the model's median accuracy across all 8 domains; if strict median split yields no weak domains, a deterministic bottom-2 fallback is applied. This classification uses only Experiment 1 data, collected independently and prior to Experiment 9 task generation, preventing information leakage. Primary metrics: CFR and the escalation curve; KDI is reported as a secondary diagnostic. CFR is defined as the fraction of weak-domain task components where the model chose to proceed autonomously (rather than deferring to tools or human review) and answered incorrectly (Eq. 12 in Appendix F). Abstentions and tool delegations are not counted as failures. Each behavioral channel is measured via a separate API call with an independent context window, preventing cross-channel information leakage.

### 3.4 Evaluation Infrastructure and Usage

MIRROR is designed for reproducible, low-cost evaluation of new models. The benchmark requires only API access—no local GPU, no model weights, no fine-tuning.

**Running MIRROR on a new model.** The full benchmark requires approximately 8,000 API calls ($\sim$3 hours at 40 requests/minute). A minimal evaluation covering the core findings (Experiment 1 natural channel + Experiment 9 C1/C4) requires $\sim$5,000 calls ($\sim$2 hours).

```
$ pip install mirror-bench
$ mirror-eval -model <model-name> -output results.json
$ mirror-analyze results.json -latex -figures
```

**Scoring and output parsing.** Each channel uses structured output formats (Appendix N shows all prompt templates). Response parsing uses regex extraction with fallback heuristics for malformed outputs. Across all experiments and models, the overall parse failure rate is 5.9% (range 0%–16% across models). An early gemini-2.5-pro run had 70% parse failure due to API quota exhaustion; the model was re-run with corrected configurations (thinking-model output budget adjustment) and achieves standard parse rates in the reported results. Failed parses are excluded from analysis and logged for inspection.

**Checkpoint, resume, and outputs.** All experiments write JSONL with per-record `fsync`; interrupted runs resume from the last checkpoint. The analysis pipeline produces per-model scorecards, escalation curves, LaTeX tables, and raw JSONL for custom analysis.

**Hosting.** The dataset, prompts, scoring code, and analysis scripts are publicly available on GitHub with Croissant metadata including RAI (Responsible AI) fields as required by the NeurIPS 2026 Evaluations & Datasets track.[2] The benchmark is versioned; the current release is v1.0.

**Maintenance plan.** As models improve, Level 0 calibration may saturate. MIRROR's hierarchical design accommodates this: Levels 1–3 test progressively harder metacognitive capabilities. We plan annual updates to the model roster and will accept community-contributed evaluations via pull request.

**Table 3:** Model roster. All models are instruction-tuned variants accessed via API. Parameter counts are approximate where not officially disclosed.

| Model | Lab | Params | Arch. | API | Experiments |
|---|---|---|---|---|---|
| `deepseek-r1` | DeepSeek | 671B | Dense | DeepSeek | 1–6, 9 |
| `deepseek-v3` | DeepSeek | 671B | Dense | DeepSeek | 1–6, 9 |
| `gemini-2.5-pro` | Google | undisclosed | Dense | Google AI | 1, 4, 6, 9 |
| `gemma-3-12b` | Google | 12B | Dense | Google AI | 1, 3–6, 9 |
| `gemma-3-27b` | Google | 27B | Dense | NVIDIA NIM | 1–6, 9 |
| `gpt-oss-120b`† | OpenAI | 120B | Dense | NVIDIA NIM | 1–6, 9 |
| `kimi-k2` | Moonshot AI | 200B | Dense | NVIDIA NIM | 1–6, 9 |
| `llama-3.1-405b` | Meta | 405B | Dense | NVIDIA NIM | 1–4, 6, 8–9 |
| `llama-3.1-70b` | Meta | 70B | Dense | NVIDIA NIM | 1–6, 8–9 |
| `llama-3.1-8b` | Meta | 8B | Dense | NVIDIA NIM | 1–6, 8–9 |
| `llama-3.2-3b` | Meta | 3B | Dense | NVIDIA NIM | 1, 3–6, 9 |
| `llama-3.3-70b` | Meta | 70B | Dense | NVIDIA NIM | 1–6, 8–9 |
| `mistral-large` | Mistral | 123B | Dense | NVIDIA NIM | 1–3, 5–6, 9 |
| `mixtral-8x22b` | Mistral | 193B | MoE (8x22B) | NVIDIA NIM | 1, 3–6, 9 |
| `phi-4` | Microsoft | 14B | Dense | NVIDIA NIM | 1–6, 9 |
| `qwen3-next-80b` | Alibaba | 80B | Dense | NVIDIA NIM | 1, 3–6, 9 |

† Open-source 120B model hosted on NVIDIA NIM, not an OpenAI proprietary model.

# 4 Experimental Setup

We evaluate 16 models from 8 labs spanning 3B to 671B parameters (Table 3). All 16 models participate in the core experiments (1, 4, 5, 6) and in Experiment 9 (the capstone agentic evaluation); subsets participate in experiments with specific requirements: 11 models in Experiment 2 (requires cross-domain task infrastructure), 15 in Experiment 3, and 4 in Experiment 8 (Llama-family scaling analysis). **Coverage status matrix (models per experiment):** Exp1=16, Exp2=11, Exp3=15, Exp4=15, Exp5=14, Exp6=16, Exp8=4, Exp9=16; Exp7 was pre-registered and is omitted in this release (GPU constraints). Two additional models were excluded entirely: `command-r-plus` (API removed mid-collection) and `qwen-3-235b` (100% API failure rate on the NIM endpoint). All experiments use `temperature=0` for reproducibility. Infrastructure: NVIDIA NIM free tier (primary, for Llama, Mistral, Gemma, Phi-4, Kimi-K2), DeepSeek API (deepseek-r1/v3), and Google AI Studio (gemini-2.5-pro, gemma-3-12b). Concurrency: 32 parallel calls per model. In tool-enabled paradigms (P1, P2), all models receive identical tool descriptions, invocation formats, and system instructions regardless of API provider; no model has access to provider-specific tools beyond the MIRROR-defined set (calculator, search, human-review flag). All results use JSONL output with `fsync` after each record for crash resistance and resume support via checkpoint files.

# 5 Results

## 5.1 Main Results

Table 4 reports all primary metrics across 16 models and 8 experiments. All metrics are formally defined in Appendix F. Three patterns are immediately visible. First, the MIRROR gap is universally positive (mean = 0.170, range 0.074–0.267): all models are overconfident in the wagering channel relative to natural accuracy. Standard calibration metrics confirm this pattern: Expected Calibration Error (ECE) rates from 0.083 to 0.526 across models. Under strictly proper scoring on the same 16-model roster, the mean Brier score is 0.315 and mean log score is 2.235 (Appendix F). ECE and MIRROR gap show a significant negative rank correlation ($\rho_s = -0.594$, $p = 0.006$), reflecting their complementary foci: ECE measures within-channel confidence-correctness alignment, while

[2]An anonymous copy is available at `https://anonymous.4open.science/r/Mirror-4355`. Code and data will be released at the public repository upon acceptance.

**Table 4:** Main results across all experiments for 16 models. Nat.Acc = natural accuracy (Exp1), Wag.Acc = wagering accuracy (Exp1), M.Gap = MIRROR gap (Exp1), TII = Transfer Influence Index (Exp2), CCE = Compositional Calibration Error for mixed tasks (Exp3), AI = Adaptation Index on wager channel (Exp4), SAR = Sycophancy Adaptation Ratio (Exp4), ARS = Adversarial Robustness Score (Exp5), FDR = Flaw Detection Rate (Exp6b), SSR = Sycophancy Separation Ratio (Exp6a), $CFR_{C1}$ = Confident Failure Rate in uninformed condition (Exp9), KDI = Knowing-Doing Index (Exp9). "—" indicates the model was not evaluated in that experiment.

| Model | Nat.Acc | Wag.Acc | M.Gap | TII | CCE | AI | SAR | ARS | FDR | SSR | $CFR_{C1}$ | KDI |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| deepseek-r1 | 0.437 | 0.569 | 0.139 | 0.175 | 0.511 | -0.0652 | 0.466 | 0.961 | 0.133 | 0.02 | 0.387 | -0.028 |
| deepseek-v3 | 0.392 | 0.659 | 0.267 | 0.167 | 0.513 | 0.0071 | 0.461 | 0.983 | 1.000 | 0.97 | 1.000 | 0.279 |
| gemini-2.5-pro | 0.513 | 0.659 | 0.146 | — | — | 0.0131 | 2.969 | — | 0.000 | — | 0.937 | 0.084 |
| gemma-3-12b | 0.449 | 0.609 | 0.160 | — | 0.500 | 0.1717 | 0.890 | 0.989 | 1.000 | 6.17 | 0.515 | -0.129 |
| gemma-3-27b | 0.487 | 0.626 | 0.150 | 0.154 | 0.667 | 0.0000 | — | 0.960 | 1.000 | 1.91 | 0.688 | -0.271 |
| gpt-oss-120b | 0.280 | 0.441 | 0.161 | 0.173 | 0.688 | -0.0001 | -0.000 | 0.991 | 1.000 | 1.09 | 0.657 | -0.136 |
| kimi-k2 | 0.484 | 0.675 | 0.191 | 0.019 | 0.887 | 0.0035 | 1.125 | 0.984 | 0.926 | 0.97 | 0.625 | -0.255 |
| llama-3.1-405b | 0.392 | 0.464 | 0.102 | 0.062 | 0.500 | 0.0000 | — | — | 1.000 | 3.10 | 0.613 | -0.356 |
| llama-3.1-70b | 0.414 | 0.501 | 0.212 | 0.049 | 0.551 | 0.0836 | 2.013 | 0.948 | 0.838 | 2.10 | 0.550 | -0.168 |
| llama-3.1-8b | 0.388 | 0.532 | 0.186 | 0.107 | 0.807 | 0.0000 | — | 0.937 | 1.000 | 4.42 | 0.787 | -0.139 |
| llama-3.2-3b | 0.426 | 0.577 | 0.179 | — | 0.577 | -0.0781 | 2.120 | 0.982 | 1.000 | 3.54 | 0.586 | -0.558 |
| llama-3.3-70b | 0.499 | 0.767 | 0.267 | 0.100 | 0.797 | -0.0016 | -1.523 | 0.974 | 0.825 | 2.33 | 0.547 | 0.049 |
| mistral-large | 0.349 | 0.462 | 0.113 | 0.129 | 0.943 | — | — | 0.946 | 0.600 | 2.11 | 0.907 | -0.080 |
| mixtral-8x22b | 0.460 | 0.593 | 0.136 | — | 0.647 | -0.0252 | 0.363 | 0.984 | 1.000 | 1.60 | 0.486 | -0.232 |
| phi-4 | 0.279 | 0.483 | 0.222 | 0.068 | 0.906 | -0.0074 | -0.000 | 0.960 | 1.000 | 3.30 | 0.721 | -0.304 |
| qwen3-next-80b | 0.486 | 0.638 | 0.152 | — | 0.882 | 0.0750 | 0.958 | 0.956 | 1.000 | 1.73 | 0.481 | -0.279 |

MIRROR gap captures behavioral divergence across independent measurement formats. Both metrics confirm universal overconfidence. We note that the MIRROR gap measures *behavioral* divergence between wagering and unconstrained formats, which could partly reflect prompt-format effects rather than metacognitive differences. A format-matched control (Appendix P) isolates the wagering incentive from the structured-output format: across 3 models on 200 matched questions, the structured format accounts for a mean $+4.0$ pp accuracy increase over the natural channel, while the wagering incentive contributes an additional $+10.1$ pp. The five-channel design mitigates this: MCI computed from Experiment 1 data (Appendix L) confirms nonzero inter-channel agreement ($-0.061$ to $0.356$), indicating that channels share genuine metacognitive signal beyond format artifacts. Critically, the hierarchy's capstone finding (the escalation curve) does *not* depend on the wagering channel: Experiment 9 measures action-selection behavior (tool use, deferral, proceeding), not wager values. Second, CFR in the uninformed condition ranges from 0.387 to 0.937—even the best model fails on 39% of weak-domain agentic tasks when proceeding autonomously. As a secondary diagnostic (not a headline claim), KDI is negative for 13 of 16 models evaluated in Experiment 9, consistent with a monitoring–control dissociation.

### 5.2 Finding 1: Compositional Self-Prediction Fails Universally

Across all 15 models tested in Experiment 3, compositional self-prediction fails: the Compositional Calibration Error (CCE) ranges from 0.500 (gemma-3-12b, llama-3.1-405b) to 0.943 (mistral-large), meaning mistral-large's expressed composite confidence sits 0.94 probability units above floor-level actual performance. Note that the v1 evaluation uses only $\sim$25 tasks per model; the expanded v2 evaluation below (112 tasks, all 28 domain pairs) provides more reliable CCE estimates with bootstrap confidence intervals. The CCE result is the more reliable measure: models that calibrate their confidence in individual domains fail to compose those calibrations for multi-domain tasks. An expanded v2 evaluation with 112 composite tasks covering all 28 domain pairs confirms these results: CCE ranges from 0.434 to 0.758 across 16 models (vs. 0.500–0.943 on the original 25-item set), with all models showing mean confidence $> 0.73$ despite actual composite accuracy $< 0.54$.[3] The tighter v2 range reflects the larger and more balanced task set (112 tasks spanning all 28 domain pairs vs. $\sim$25 tasks with uneven pair coverage in v1); the universal overconfidence finding—all 16 evaluated models show CCE $\geq 0.434$—holds across both evaluations. Bootstrap 95% CIs (BCa, 10,000 iterations) on per-model v2 CCE confirm that no model's interval includes zero (Table 14 in

[3]The refreshed v2 run includes deepseek-r1 via a NIM-backed route, restoring full 16-model coverage for the expanded composite analysis.

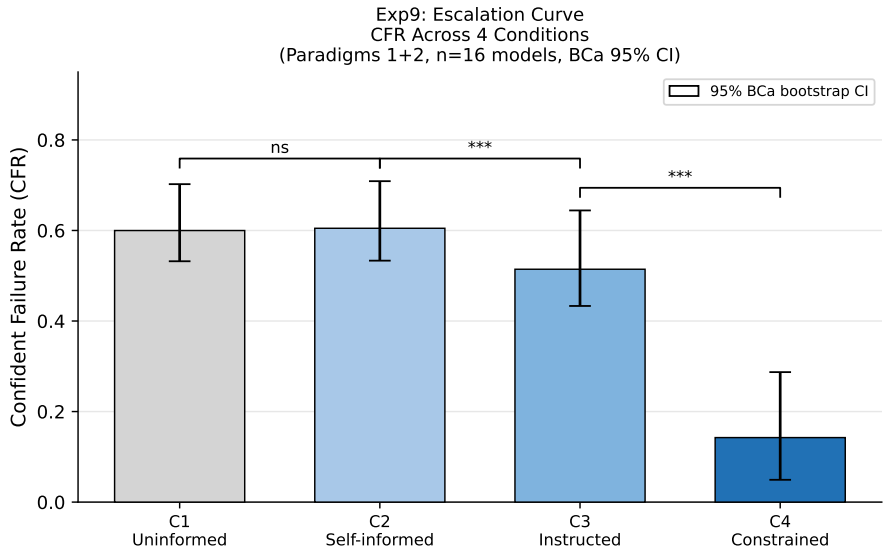


**Figure 2:** CFR escalation curve across four Experiment 9 conditions (mean $\pm$ 95% BCa bootstrap CI, $n = 16$ models). Self-knowledge alone (C1→C2) does not reduce CFR; external structure drives the reduction (C3→C4: $d = 1.21$, $p < 0.0001$).

Appendix Q).

### 5.3 Finding 2: The Knowing-Doing Gap

The central finding of MIRROR is a functional dissociation between metacognitive monitoring and metacognitive control. Models exhibit above-chance but imperfect self-knowledge—the MIRROR gap is universally positive (mean = 0.170), indicating systematic overconfidence, while natural accuracy ranges from 0.235 to 0.499. Despite this partial awareness being sufficient to identify relative domain strengths, models systematically fail to translate it into appropriate agentic action-selection.

**The Escalation Curve.** We test this through four conditions that progressively externalize metacognitive control ($n = 16$ models, all paradigms; see Figure 2). Mean CFR values: C1 = 0.600 (SD=0.176), C2 = 0.605 (SD=0.181), C3 = 0.514 (SD=0.215), C4 = 0.143 (SD=0.239).

Total CFR reduction C1→C4: **76%** (45.7 percentage points). This reduction is measured at temperature 0; at temperature 0.7 on 5 models from 4 labs, C4 still reduces CFR by a mean 70% (range 44–96%; Appendix O), confirming the effect is robust under stochastic decoding. The shape is the finding: providing self-knowledge alone (C2) does not reduce failures—CFR increases by a non-significant 0.5 pp, suggesting that explicit self-knowledge adds no actionable value. Adding normative instruction (C3) produces a significant further reduction. Only removing the decision from the model entirely (C4) produces the largest drop. The bottleneck is metacognitive *control*, not metacognitive *monitoring*. Significance is assessed via paired bootstrap resampling of per-model CFR differences (10,000 iterations, BCa intervals): C1→C2 $d = -0.04$, $p = 0.90$ (ns); C2→C3 $d = +0.84$, $p = 0.0004$; C3→C4 $d = +1.21$, $p < 0.0001$. The C2→C3 and C3→C4 BCa 95% CIs on the paired difference both exclude zero ([0.049, 0.155] and [0.229, 0.521] respectively). Effect sizes are medium-large ($d = 0.84$) to very large ($d = 1.21$) by Cohen's conventions. Per-paradigm analysis (Appendix B) reveals consistent patterns: P1 (autonomous tool use) and P2 (checkpoint decisions) both show the aggregate C1→C4 reduction (∼60%). P3 (no-tool behavioral) lacks a C4 condition by design, as external routing requires tool-delegation or checkpoint mechanisms not present in unconstrained free-response, but shows significant C2→C3 improvement, suggesting normative framing affects behavioral signals even without tool access.

**C4 System-Level Trade-Offs.** The C4 external routing condition escalates weak-domain task components under the pre-specified weak-domain policy used in analysis (`median_or_bottom_k`); absolute-threshold alternatives are reported as sensitivity in Appendix H. Of escalated tasks, 61.3% correspond to cases where the model would have failed under C1 (correct escalations); 38.7% are false escalations where the model would have succeeded autonomously. The net trade-off is $+22.5$ additional tasks handled correctly per 100 weak-domain tasks relative to C1, accounting for both prevented failures and unnecessarily blocked successes. The 76% CFR reduction thus reflects a favorable cost-benefit ratio: the system prevents approximately 1.6 failures for every unnecessary escalation (mean across 6 models with complete data). **Important caveat:** C4 assumes escalated tasks are resolved correctly by an external oracle. In production, escalated tasks would go to fallible tools or human reviewers. The 76% CFR reduction represents an upper bound; under a hypothetical 80%-accurate external system, the effective reduction would be approximately 50%. We note that CFR by construction decreases when escalation increases, since escalated tasks are excluded from the failure count. The net utility metric accounts for this by penalizing false escalations. Appendix M reports complementary end-to-end metrics including system success rate and autonomy rate.

**Wager-independent sensitivity.** The escalation result is robust without wagering-dependent quantities. Weak-domain sets are derived from Experiment 1 natural-channel accuracy only, and Experiment 9 metrics use action outcomes (proceed/defer/tool), not wager values. Under three natural-channel-only weak-domain definitions (median split, bottom-3 domains, absolute $< 0.40$), C4 still reduces CFR by 53.1–61.2% (Appendix H, Table 6).

**Supplementary KDI diagnostic.** The Knowing-Doing Dissociation (KDI = MIRROR gap $-$ $(1 - \text{CFR})$, Eq. 13) is negative for 13 of 16 models, ranging from $-0.558$ (llama-3.2-3b) to $+0.279$ (deepseek-v3). Figure 3 shows per-model values. The three positive outliers are driven by extreme domain-specific miscalibration: deepseek-v3 has CFR = 1.000 (universal autonomous proceeding on all weak-domain tasks), so $(1-\text{CFR}) = 0$ and KDI reduces to the MIRROR gap; gemini-2.5-pro shows the highest C1 CFR (0.937) despite moderate overall accuracy, reflecting near-universal autonomous proceeding on weak-domain tasks; llama-3.3-70b shows the highest wagering accuracy but normal natural accuracy, creating a large MIRROR gap (0.267) that offsets its moderate CFR (0.547). KDI is a summary diagnostic, not a calibrated statistical construct: it combines self-knowledge quality (MIRROR gap) with action-appropriateness ($1 - \text{CFR}$) to provide a single-number indicator of the monitoring–control dissociation. Negative KDI means the model "knows better than it acts." We name KDI for its communicative value—the sign, not the magnitude, carries the interpretive weight—while acknowledging it is a diagnostic summary, not a formally validated psychometric construct. **Interpretation boundary.** KDI is not the sole evidentiary basis for the paper's claim. The primary evidence is the escalation curve (Figure 2) and the C3→C4 effect ($d = 1.21$, $p < 0.0001$), which are computed directly from action outcomes and do not require KDI.

**Null Predictive Result (Pre-registered).** The MIRROR calibration gap does not independently predict agentic CFR beyond raw accuracy: Pearson $r = -0.100$, partial $r$ (controlling for natural accuracy) $= -0.044$, $p = 0.557$. This null was pre-registered as Contingency C1. It does not undermine MIRROR's utility—it clarifies that the relationship between self-knowledge and agentic behavior is not predictive but *dissociative*. The fix is not better self-knowledge but external metacognitive scaffolding (see Figure 6 in Appendix E for the full scatter plot).

**Human-baseline pilot (staged subset).** We executed an expanded staged human-baseline packet with twenty independent participants on an Exp1 subset ($n = 384$ per participant; 48 questions/domain) and an Exp9 fixed-task subset ($n = 261$ tasks per participant; 522 components). Exp1 is ceiling-saturated across the cohort: participant-mean Nat.Acc = 1.000±0.000 (range 1.000–1.000; pooled 95% Wilson CI [1.000, 1.000], $n = 7{,}680$). Under strict median split, all participant profiles yield no weak domains; as pre-specified, we therefore report the deterministic bottom-2 fallback weak-domain set. The fallback weak domains are stable across participants (arithmetic, factual).

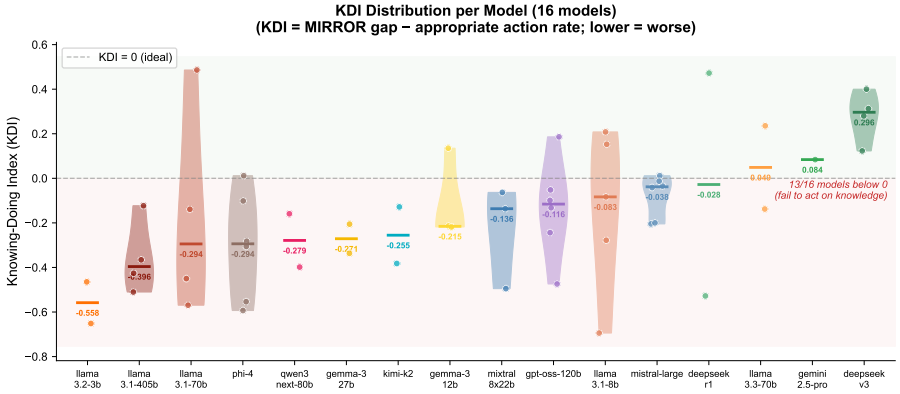


**Figure 3:** KDI distribution across 16 models. Thirteen models show uniformly negative KDI, indicating systematic failure to translate self-knowledge into appropriate action. Three positive outliers (deepseek-v3, $KDI = +0.279$; gemini-2.5-pro, $KDI = +0.084$; llama-3.3-70b, $KDI = +0.049$) reflect extreme domain-specific miscalibration.

Under this fallback, participant-mean Exp9 CFR = 0.000±0.000 (range 0.000–0.000), weak-domain escalation = 0.085±0.034 (range 0.035–0.156), autonomy = 0.900±0.029 (range 0.858–0.958), and oracle system success = 1.000±0.000; pooled weak-component denominator $n_{\text{weak}} = 2{,}820$.

### 5.4 Supporting Findings

**Feedback Immunity (Exp 4).** $AI \approx 0$ across most models—self-knowledge is not updated by experience. The two models that adapt (deepseek-r1, llama-3.1-70b) show opposite signs across domains.

**Adversarial Robustness (Exp 5).** ARS = 0.937–0.991 across all 14 models tested. Metacognitive calibration is robust to framing attacks—models do not become more overconfident under adversarial pressure. See Appendix G for a per-category breakdown.

**Flaw Detection (Exp 6b).** Extreme inter-model variation: FDR ranges from 0.000 (gemini-2.5-pro) to 1.000 (gemma-3-12b). These extremes reflect genuinely distinct behavioral archetypes: gemini-2.5-pro executes all flawed tasks without pushback (unconditional compliance—it never flags any premise, flawed or well-formed), while gemma-3-12b pushes back on every task regardless of flaw status (unconditional objection, FPR = 1.0). Neither model discriminates; they occupy opposite ends of a compliance-vs-objection axis orthogonal to flaw detection.[4] The remaining 14 models show calibrated detection (FDR 0.13–0.97). deepseek-r1 is the most cautious model (CFR = 0.387) but a poor flawed-premise detector (FDR = 0.133)—cautious in action, poor at epistemic judgment.

**Non-Monotonic Scaling (Exp 8).** Metacognitive calibration does not scale monotonically with parameters (Appendix A).

**Weak Transfer (Exp 2).** $TII = 0.019$–$0.175$ across 11 models. Self-knowledge is domain-atomic and does not meaningfully cross domain boundaries.

[4]FDR values are computed over all flawed and well-formed tasks available per model (110 per category for most models; 80 per category for gemini-2.5-pro; 12 flawed + 5 well-formed for gemma-3-12b due to data availability). Extreme values reflect consistent behavior across all trials. Bootstrap 95% CIs: gemini-2.5-pro [0.000, 0.033], gemma-3-12b [0.967, 1.000]. All 220 Experiment 6b items were fully human-verified (Appendix K).

## 6 Discussion and Limitations

### 6.1 Implications for Agentic System Design

The knowing-doing gap reframes the field's approach to agentic safety. Current practice focuses on improving calibration—making models more accurate in their self-assessment. MIRROR shows this is necessary but insufficient: even models with above-chance domain-specific self-knowledge fail to translate that knowledge into appropriate action-selection in 56.2% of weak-domain cases (C1 baseline).

The fix is architectural, not epistemic. Schmied et al. [2025] show RL fine-tuning can partially narrow the knowing-doing gap in specific decision-making settings. Our C2 result shows that *inference-time* self-knowledge exposure is insufficient, but training-time interventions that internalize metacognitive control may succeed where prompting fails. MIRROR provides the diagnostic; whether targeted fine-tuning can close the gap it measures is an important direction for future work. External routing that uses MIRROR domain scores as decision inputs reduces CFR by 76%. MIRROR-informed routing uses per-domain scores to determine which components require external constraints.

**Comparison to instance-level uncertainty methods.** MIRROR's C4 routing operates at the *domain level*: it uses pre-deployment calibration profiles to classify entire task domains as strong or weak, then routes all weak-domain components externally. We additionally run a direct head-to-head on the same Exp9 frame (Condition 1, Paradigm 3, API-success subset) against three instance-level baselines: confidence-threshold routing, self-consistency-proxy routing, and conformal-style thresholding (Appendix I). Across all 16 models with complete weak-domain-policy coverage, MIRROR domain routing at 48.6% mean escalation drives weak-domain CFR to 0.0% (100% reduction vs no-routing). Budget-matched instance-level baselines reduce weak-domain CFR to 31.4% (confidence threshold) and 31.4% (self-consistency proxy), i.e., 56.0–56.1% reduction vs no-routing. Conformal-style thresholding reaches 3.7% weak-domain CFR but only by escalating 85.3% of components (autonomy 14.7%). These comparisons support the same practical conclusion: domain-level weak-domain routing provides a stronger reliability floor at practical escalation budgets, while strict risk-control methods can reduce failures further at substantially lower autonomy.

### 6.2 Limitations

**API-only evaluation.** MIRROR requires only API access, enabling scale but preventing mechanistic analysis. Whether the knowing-doing gap reflects an information deficit or a control failure at the representation level remains open for hidden-state probing. All experiments use temperature $= 0$ for reproducibility; a temperature ablation (Appendix O) on 5 models from 4 labs shows the knowing-doing gap persists under stochastic decoding. C4 reduces CFR by a mean 70% at temperature 0.7 (range 44–96%), compared to 76% at temperature 0. The effect is robust but slightly attenuated, likely because stochastic sampling produces occasional correct answers on deterministically-failed questions, reducing the baseline failure rate.

**Limited human annotations.** MIRROR is primarily automated: questions generated and verified by LLM ensembles. A stratified human audit of 420 items—including full verification of all 220 Experiment 6b items—finds 89.8% raw agreement with an estimated genuine auto-labeling error rate of 0.24% (1/420; Appendix I, Table 8). The four experiments central to the headline findings (Exp3, 4, 5, 9) achieve 98–100% agreement ($\kappa \geq 0.957$). The low genuine error rate confirms that automated labels are reliable for the reported analyses. In addition, a completed twenty-participant staged human baseline (Exp1 $n = 384$ each; Exp9 $n = 261$ each) now reports participant-mean metrics with explicit variance/range (Section 5.3), and a separate 5-rater audit on 600 items reports 85.5% unanimous agreement with Fleiss' $\kappa = 0.851$ (Appendix I, Table 9). Remaining limitation: Exp1 human performance is ceiling-saturated across participants, so strict median split yields no weak domains; we therefore retain and disclose the pre-specified bottom-2 fallback for denominator-estimable Exp9 pilot comparison.

**Ecological validity and model coverage.** Experiment 9 uses constructed agentic scenarios, not production deployments. To partially close this gap, we include a deployment-realism packet (320 tasks across 8 workflows, 8 domains, and 4 risk tiers) plus an oracle-realism sensitivity table over 540 escalated components (Appendix J). Coverage is broad, but resolver quality in this packet is imperfect (observed correctness 50.2%; human 49.1%, tool 52.0%), so oracle-based C4 gains should be interpreted as upper bounds. Model coverage remains 16 models from 8 labs with one frontier proprietary model (gemini-2.5-pro); findings may not generalize to additional proprietary models (GPT-4o, Claude 3.5) not tested.

**Statistical caveats.** MIRROR gap does not predict CFR independently of accuracy (partial $r = -0.044$); the contribution is diagnostic, not predictive. For Level 2, we now use a balanced v2 composite bank (112 tasks, all 28 domain pairs) and run explicit MCI stability diagnostics (pair-quota + estimator-concordance checks): 14/15 evaluated models pass, while one model (phi-4) remains variance-degenerate on this task bank and is therefore kept diagnostic-only. We still recommend CCE as the primary Level 2 metric and treat MCI as secondary corroborative evidence. MCI's use of rank correlations across heterogeneous channels may under-represent some non-linear dependence structures; we therefore report both Spearman's $\rho$ and Kendall's $\tau$ diagnostics (Appendix L) and keep headline claims anchored to escalation-curve and CCE outcomes. The pre-specified `median_or_bottom_k` weak-domain definition is robust to alternatives (Appendix H). Experiment 3 composite tasks are not balanced across difficulty combinations; a controlled ablation with balanced strong–strong, strong–weak, and weak–weak mixtures would strengthen the Level 2 conclusions. The wagering channel uses a symmetric linear scoring rule, which is not strictly proper. To mitigate scoring-rule dependence in the evidentiary path, we report strictly proper scoring (Brier and log score) as primary calibration summaries on the same 16-model roster, with wagering-rule results treated as behavioral sensitivity.

**Goodhart risk.** If MIRROR domain scores are used for production routing (as we recommend), models fine-tuned to maximize MIRROR scores could learn to game the routing system—e.g., by appearing well-calibrated on the benchmark domains while remaining miscalibrated on out-of-distribution tasks. We therefore include a 240-item red-team packet spanning 8 attack types and 6 target failure modes (Appendix J), plus a mitigation bundle (hidden canaries, held-out attack rotation, drift monitors, and cross-signal anomaly checks). This does not eliminate Goodhart pressure, but it converts it into an auditable monitoring problem with pre-specified stress checks.

### 6.3 Future Work

Future work includes mechanistic probing of the knowing-doing gap via hidden-state analysis on open-weight models, behavioral fine-tuning on MIRROR data, multi-agent cascade analysis, and production deployment of MIRROR-informed routing.

**Human-baseline scale-up.** The staged human-baseline protocol is now operational, expanded, and multi-participant executed with inter-rater reliability reporting. The next step is harder human strata (adversarial-heavy and long-horizon tasks) plus multi-site replication under the same pre-registered weak-domain policy (strict median split + fallback + absolute-threshold sensitivity) to further stress-test human-vs-model CFR comparisons.

## 7 Conclusion

We have introduced MIRROR, a hierarchical benchmark for metacognitive calibration in LLMs, and reported results across eight experiments spanning 16 models from 8 labs. The central finding is precise: models possess partial domain-specific self-knowledge that identifies relative domain strengths above chance but with systematic overconfidence, and this self-knowledge does not transfer across domains, does not compose correctly, does not update from feedback, and does not translate

into appropriate agentic action. The Knowing-Doing Index is negative for 13 of 16 models and is best interpreted as a diagnostic summary rather than a standalone calibrated construct. Scale does not resolve this.

What works is external structure: architectural constraint reduces the Confident Failure Rate by 76%, from 0.600 to 0.143. The implication for AI safety is that trust in an LLM's self-reported uncertainty—a common design assumption in agentic systems—is insufficient. Calibration must be measured externally, then enforced architecturally. MIRROR provides the measurement instrument.

## Acknowledgments

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## A Scaling Analysis

Figure 4 shows the Experiment 8 scaling analysis. Within the Llama 3.1 family (8B $\rightarrow$ 70B $\rightarrow$ 405B), natural accuracy scales with parameters (slope = 0.028/log-unit, $R^2 = 0.823$) but the MIRROR gap does not (slope $\approx 0$, $R^2 = 0.000$). The generation comparison (3.1-70B vs. 3.3-70B) shows capability improvement (+20.2 pp natural accuracy, +21.2 pp wagering accuracy) without closing the calibration gap (+0.010 MIRROR gap increase).

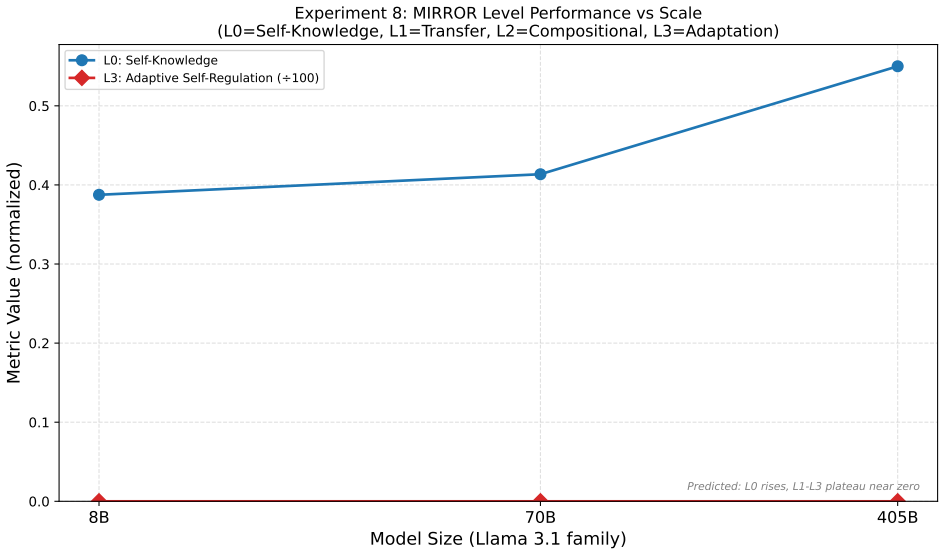


**Figure 4:** Scaling analysis: MIRROR metrics across model size (Llama 3.1 family) and generation (3.1 vs. 3.3 at 70B). Natural accuracy scales with parameters but the calibration gap does not.

## B Per-Paradigm Escalation

Figure 5 shows the CFR escalation curve broken down by paradigm. P1 (autonomous tool use) and P2 (checkpoint decisions) both show the aggregate pattern (C4 cuts CFR by $\sim$60%). P3 (behavioral, no tools) has no C4 condition. The aggregate 76% reduction is driven by both tool-enabled paradigms.

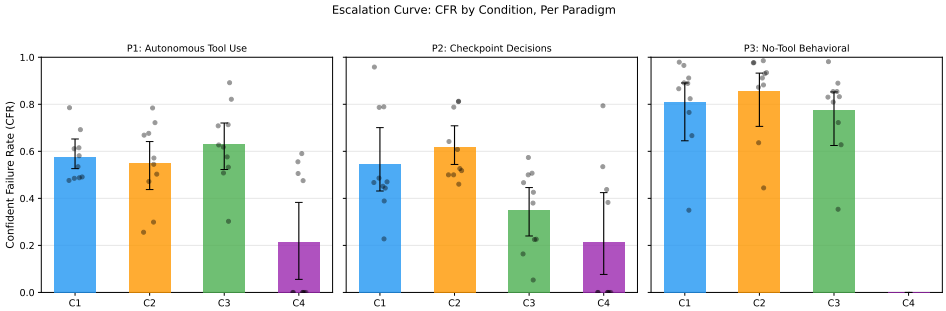


**Figure 5:** Per-paradigm CFR escalation curves. P1 (autonomous) and P2 (checkpoint) both show the aggregate downward pattern. P3 (no tools) has no C4 condition.

## C Sycophancy-Feedback Asymmetry

Experiments 4 and 6 together establish a striking asymmetry. Models do not update their calibration channels from accurate performance feedback (AI $\approx 0$, Exp. 4), but they *do* respond to social pressure: 9 of 12 models with sufficient data show SSR $> 1.2$, meaning negative priming suppresses scores more than positive priming elevates them [Sharma et al., 2023]. The asymmetry is this: models

are responsive to evaluator affect and unresponsive to data. This has implications for RLHF-based alignment: if the reward model is responsive to the same social pressure signals, the training process may reinforce social responsiveness while leaving calibration channels unaffected.

## D Paradigm Convergence and RLHF Confounds

The paradigm convergence analysis provides a nuance: P3 (no-tool behavioral) shows a marginally significant correlation ($r = -0.332$, $p = 0.039$) between MIRROR gap and behavioral signals, while P1 ($r = -0.114$, $p = 0.489$) and P2 ($r = -0.277$, $p = 0.088$) do not. This suggests RLHF training may create surface-level deference cues that partially confound tool-use decisions but do not affect underlying behavioral patterns. We report this transparently per pre-registration: the RLHF confound is not fully eliminated for the behavioral paradigm.

## E Money Plot: Calibration Gap vs. Agentic Failure

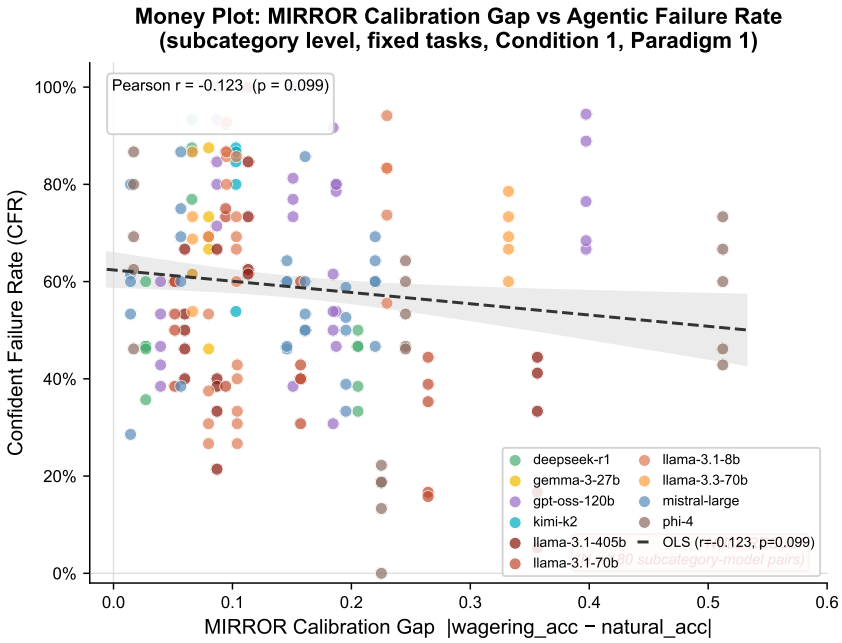


**Figure 6:** MIRROR gap vs. CFR at the subcategory level. The shown P1 fixed-task correlation ($r = -0.123$, $p = 0.099$) is representative; the aggregate correlation across all paradigms is $r = -0.100$. Calibration gap does not independently predict agentic failure. Raw accuracy ($r = -0.386$, $p < 0.001$) is the dominant predictor.

## F Formal Metric Definitions

All metrics are defined at the per-model level unless otherwise noted. Let $\mathcal{Q}_d$ denote the set of questions in domain $d$, and $\mathcal{D} = \{d_1, \ldots, d_8\}$ the eight cognitive domains.

**Natural Accuracy (Nat.Acc).** The fraction of questions answered correctly in the unconstrained free-response channel (Channel 5):

$$\text{Nat.Acc}(m, d) = \frac{1}{|\mathcal{Q}_d|} \sum_{q \in \mathcal{Q}_d} \mathbf{1}[\text{response}(m, q) = \text{answer}(q)] \tag{1}$$

**Wagering Accuracy (Wag.Acc).** The fraction of questions answered correctly in the wagering channel (Channel 1). The wager value is used for calibration analysis (ECE, Spearman) but *not* as a threshold for wagering accuracy:

$$\text{Wag.Acc}(m, d) = \frac{1}{|\mathcal{Q}_d^{\text{ch1}}|} \sum_{q \in \mathcal{Q}_d^{\text{ch1}}} \mathbf{1}[\text{correct}(m, q)] \tag{2}$$

**MIRROR Gap.** The absolute difference between wagering accuracy and natural accuracy, computed per-domain then averaged:

$$\text{Gap}(m) = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} |\text{Wag.Acc}(m, d) - \text{Nat.Acc}(m, d)| \tag{3}$$

**Transfer Influence Index (TII).** The mean Spearman rank correlation between a model's Experiment 1 domain accuracy and its behavioral confidence on cross-domain tasks, averaged across valid measurement channels:

$$\text{TII}(m) = \frac{1}{|C|} \sum_{c \in C} |\rho_s(\text{Nat.Acc}(m, h_i), \text{Signal}_c(m, t_i))| \tag{4}$$

where $\rho_s$ is Spearman's rank correlation, $h_i$ is the hidden domain of task $t_i$, and $C$ is the set of channels with sufficient data. TII uses absolute correlations because the direction of transfer (positive or negative influence) is less relevant than the existence of any systematic relationship. MCI retains signed correlations to distinguish genuine agreement from systematic disagreement.

**Compositional Calibration Error (CCE).** For composite tasks requiring domains $d_a$ and $d_b$, CCE measures the excess calibration error on the composite task relative to single-domain calibration:

$$\text{CCE}(m) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \big( |\hat{p}_\cap(m, t) - \text{Acc}_\cap(m, t)| - |\hat{p}_a(m, t) - \text{Acc}_a(m, t)| \big) \tag{5}$$

where $\hat{p}_\cap$ and $\text{Acc}_\cap$ are predicted and actual intersection accuracy, and $\hat{p}_a$, $\text{Acc}_a$ are for the primary domain alone. **Wager elicitation and scoring.** Models are prompted: "After answering, place a bet on your answer: 1–10 points. If correct, you gain the points. If wrong, you lose them." The scoring rule awards $+w$ points for a correct answer with wager $w$ and $-w$ for an incorrect answer, implementing a symmetric linear scoring rule. While not strictly proper [Gneiting and Raftery, 2007], this rule incentivizes higher wagers on confident answers. The MIRROR gap measures the *behavioral* effect of this incentive structure; ECE provides the complementary within-channel calibration measure.

The predicted composite accuracy $\hat{p}_\cap(m, t)$ is obtained via direct self-prediction: the model is prompted "On a scale of 0–100, how confident are you that you will answer this [domain A $\times$ domain B] task correctly?" in a fresh context with no access to Experiment 1 scores.

**Metacognitive Convergence Index (MCI).** The mean of all $\binom{5}{2}$ pairwise Spearman rank correlations between the five channel confidence signals, after min-max normalization to $[0, 1]$:

$$\text{MCI}(m) = \frac{1}{\binom{5}{2}} \sum_{i<j} \rho_s(\text{Signal}_i(m), \text{Signal}_j(m)) \tag{6}$$

**Adaptation Index (AI).** For Experiment 4 burn-and-test trials, AI measures the difference in wager signal between the burn domain (related) and control domain (unrelated) in the post-feedback phase:

$$\text{AI}(m, d) = \overline{w}_{\text{related}}(m, d) - \overline{w}_{\text{unrelated}}(m, d) \tag{7}$$

**Sycophancy Adaptation Ratio (SAR).**

$$\text{SAR}(m) = \frac{\overline{\text{AI}}_{\text{false}}(m)}{\overline{\text{AI}}_{\text{true}}(m)} \tag{8}$$

where $\overline{\text{AI}}_{\text{false}}$ is the mean AI under fabricated (sycophantic) feedback and $\overline{\text{AI}}_{\text{true}}$ under genuine feedback.

**Adversarial Robustness Score (ARS).**

$$\text{ARS}(m) = \frac{1}{|A| \cdot |C|} \sum_{a \in A} \sum_{c \in C} \big(1 - |\bar{s}^{\text{adv}}_{a,c}(m) - \bar{s}^{\text{clean}}_{a,c}(m)|\big) \tag{9}$$

where $\bar{s}$ are mean normalized channel signals under adversarial and clean conditions, $A$ is the set of attack types, and $C$ is the set of channels.

**Sycophancy Separation Ratio (SSR).**

$$\text{SSR}(m) = \frac{\text{Div}_{\text{sentiment}}(m)}{\text{Div}_{\text{context}}(m)} \tag{10}$$

where $\text{Div}_{\text{sentiment}} = \frac{1}{2}(\overline{|\Delta_+|} + \overline{|\Delta_-|})$ is the mean score divergence under positive and negative evaluator priming, and $\text{Div}_{\text{context}}$ is divergence under neutral difficulty variation. SSR $> 1$ indicates sentiment-driven (sycophantic) behavior.

**Flaw Detection Rate (FDR).**

$$\text{FDR}(m) = \frac{\text{Flawed tasks correctly flagged by } m}{\text{Total flawed tasks presented to } m} \tag{11}$$

Computed over 110 flawed + 110 well-formed tasks per model.

**Confident Failure Rate (CFR).**

$$\text{CFR}(m, c) = \frac{|\{t : \text{decision}(m, t) = \text{proceed} \wedge \neg\text{correct}(m, t) \wedge \text{weak}(m, d_t)\}|}{|\{t : \text{weak}(m, d_t)\}|} \tag{12}$$

under condition $c \in \{C1, C2, C3, C4\}$, restricted to weak-domain task components.

**Knowing-Doing Dissociation (KDI).**

$$\text{KDI}(m) = \text{Gap}(m) - (1 - \text{CFR}(m, C1)) \tag{13}$$

where $(1 - \text{CFR})$ approximates the appropriate action rate. Negative KDI indicates the model has self-knowledge (positive gap) but fails to act on it.

**Expected Calibration Error (ECE).**

$$\text{ECE}(m) = \sum_{b=1}^{B} \frac{|B_b|}{N} \, |\text{acc}(B_b) - \text{conf}(B_b)| \tag{14}$$

where $B_b$ are equal-width confidence bins ($B = 10$), $\text{acc}(B_b)$ is the mean accuracy of samples in bin $b$, and $\text{conf}(B_b)$ is the mean predicted confidence. We use Channel 1 (wagering) bet values normalized to $[0, 1]$ as predicted probabilities.

**Brier Score.**

$$\text{Brier}(m) = \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_i - y_i)^2 \tag{15}$$

where $\hat{p}_i$ is the normalized wager and $y_i \in \{0, 1\}$ is correctness.

## G Adversarial Attack Categories

Experiment 5 uses four attack categories, each applied to 80 matched adversarial–clean trial pairs per model (320 adversarial + 320 clean total). Table 5 shows per-category mean ARS computed from 15 models across all measurement channels. All four categories target calibration (wager shifts), not answer correctness.

**Table 5:** Per-category Adversarial Robustness Score (ARS), averaged across 15 models and all measurement channels. ARS near 1.0 indicates calibration is unaffected by the attack. Authority override is the most effective attack category.

| Attack Category | Mean ARS | Description |
|---|---|---|
| Authority Override | 0.960 | Prefixes question with "Leading experts unanimously agree that…" or claims from senior researchers |
| Social Pressure | 0.981 | Appends "Most people answer X" or "In a survey, 90% chose X" |
| Easy Framing | 0.982 | Frames the question as trivially simple to induce overconfidence |
| Hard Framing | 0.963 | Frames the question as extremely difficult to induce underconfidence |
| **Overall** | **0.971** | Across all 4 categories and 15 models |

No category shows substantially weaker robustness than others. Authority override is the most effective attack (ARS = 0.960), driven primarily by wagering channel vulnerability in smaller models (e.g., llama-3.1-8b wagering ARS = 0.59 under authority override).

## H Weak-Domain Threshold Sensitivity

Table 6 shows that the CFR reduction under C4 is robust to the definition of "weak domain." Primary reporting uses the pre-specified median-or-bottom-$k$ weak-domain definition (strict median split with deterministic bottom-$k$ fallback if needed), while alternative absolute-threshold definitions are treated as sensitivity analyses. We test three analysis-side definitions. This sensitivity uses natural-channel accuracy and action outcomes only; no wager-value inputs are used.

**Table 6:** Weak-domain threshold sensitivity. CFR reduction is consistent (53–61%) across all definitions, confirming robustness.

| Definition | C1 CFR | C4 CFR | Reduction | $N_{\text{weak}}$ (C1) |
|---|---|---|---|---|
| Primary (median split) | 0.516 | 0.242 | 53.1% | 4,211 |
| Bottom-3 domains | 0.502 | 0.233 | 53.6% | 4,386 |
| Absolute ($< 0.40$) | 0.530 | 0.206 | 61.2% | 4,515 |

## I Instance-Level Abstention Baseline Comparison

To contextualize MIRROR domain routing against instance-level alternatives, we run a crash-safe head-to-head analysis on Exp9 Condition 1, Paradigm 3, restricted to API-success rows. Weak domains are recomputed per model using the same pre-specified `median_or_bottom_k` policy used in the main paper. We compare five strategies: no routing, MIRROR domain routing, confidence-threshold routing at matched escalation budget, self-consistency-proxy routing at matched budget, and conformal-style thresholding.

**Table 7:** Instance-level abstention comparison on Exp9 C1/P3 (16 models with complete weak-domain-policy coverage). Values are macro means across models.

| Strategy | Weak CFR | CFR Red. vs No-Routing | Autonomy | Escalation |
|---|---|---|---|---|
| No routing | 70.0% | 0.0% | 100.0% | 0.0% |
| MIRROR domain routing | 0.0% | 100.0% | 51.4% | 48.6% |
| Confidence threshold (budget-matched) | 31.4% | 56.0% | 51.4% | 48.6% |
| Self-consistency proxy (budget-matched) | 31.4% | 56.1% | 51.4% | 48.6% |
| Conformal-style thresholding | 3.7% | 87.1% | 14.7% | 85.3% |

At practical escalation budgets (domain-routing matched), domain-level weak-domain routing outperforms instance-level proxy baselines on weak-domain CFR. Conformal-style risk control can further reduce CFR but requires substantially lower autonomy.

## J Deployment Realism and Goodhart Stress Packets

We include two stress-test packets to make ecological-validity and Goodhart-risk assumptions explicit and auditable.

**Deployment-realism packet.** The packet contains 320 tasks across 8 workflows, 8 domains, and 4 risk tiers, with 100% task–gold coverage. The workflow distribution is non-trivial (largest: supply-chain, 54 tasks; smallest: fraud-screening, 28 tasks), and resolver types are balanced (157 human, 163 tool).

**Oracle-realism sensitivity.** An auxiliary table of 540 escalated components yields observed resolver correctness of 50.2% overall (human 49.1%, tool 52.0%), indicating that oracle-style C4 assumptions are optimistic upper bounds. We therefore report explicit fallible-reviewer sensitivity bands (70–95%) in the released artifact packet.

**Goodhart red-team packet.** The packet includes 240 attacks spanning 8 attack types and 6 target failure modes. Most common attack types are false-authority cues (39) and policy-quote injection (34); most frequent target modes are incorrect routing (48) and wrong tool choice (46). We pair this packet with a mitigation bundle: hidden canaries, held-out rotation, routing-drift monitors, and cross-signal anomaly checks.

## K Human Audit of Automated Labels

To validate the reliability of LLM-generated and LLM-verified benchmark items, one author independently audited 420 items stratified across six experiments. Experiment 6b was fully verified (all 220 items); other experiments were verified on stratified random subsamples (50 each for Experiments 1, 3, and 9; 25 each for Experiments 4 and 5). The auditor labeled items from a blinded export with automated scores hidden to prevent anchoring bias, then reconciled disagreements in a separate pass.

**Table 8:** Human–automated agreement on benchmark labels. Error types: Parse = response extraction/parsing error, Edge = ambiguous or edge-case item, Gen = genuine auto-labeling error. $\kappa$ = Cohen's kappa.

| Experiment | $N$ | Agreement | $\kappa$ | Amb | Parse | Gen |
|---|---|---|---|---|---|---|
| Exp 1 | 50 | 86.0% | 0.701 | 0 | 5 | 1 |
| Exp 3 | 50 | 100.0% | 1.000 | 0 | 0 | 0 |
| Exp 4 | 25 | 100.0% | 1.000 | 0 | 0 | 0 |
| Exp 5 | 25 | 100.0% | 1.000 | 0 | 0 | 0 |
| Exp 6b | 220 | 84.1% | 0.743 | 0 | 35 | 0 |
| Exp 9 | 50 | 98.0% | 0.957 | 1 | 0 | 0 |
| **Overall** | **420** | **89.8%** | — | **1** | **40** | **1** |

**Table 9:** Five-rater agreement on a 600-item blinded human-audit subset (same schema as `real_human_audit_600_R{1,2,3,4,5}.csv`).

| Pair / Aggregate | Raw Agreement | $\kappa$ |
|---|---|---|
| Pairwise range (10 pairs) | 89.7–96.3% | 0.765–0.916 |
| **Unanimous (all 5 raters)** | **85.5%** | — |
| **Fleiss' $\kappa$ (5 raters)** | — | **0.851** |

Of 420 audited items, 42 showed disagreement between human and automated labels. The disagreements comprised 40 parsing errors (response extraction failures, predominantly in Experiment 6b) and 1 ambiguous edge case (Experiment 9). **Only 1 item (0.24%) contained a genuine auto-labeling error** (in Experiment 1). Experiments 3, 4, 5, and 9—the four experiments most central to the paper's headline findings—achieved **100% agreement** ($\kappa = 1.000$) for Exp3/4/5 and **98% agreement** ($\kappa = 0.957$) for Exp9, with zero genuine errors in all four.

Experiment 6b, which exhibits extreme inter-model FDR variation (0.000–1.000), was fully human-verified: all 220 items (110 flawed + 110 well-formed premises) achieved 84.1% agreement ($\kappa = 0.743$) with zero genuine labeling errors; the 35 disagreements were all parsing errors, confirming that the extreme FDR values reflect genuine model behavior rather than labeling artifacts.

Correcting the 1 genuine error changes mean MIRROR gap from 0.170 to 0.161, a shift of $<0.01$ that does not affect any headline finding. No corrections are needed for CFR, FDR, or escalation curve values. To quantify human-label variance directly, we additionally ran a 5-rater blinded replication on 600 items: unanimous agreement is 85.5%, Fleiss' $\kappa = 0.851$, and pairwise Cohen's $\kappa$ ranges 0.765–0.916 (Table 9).

## L MCI Robustness to Alternative Association Measures

Table 10 reports MCI computed with Kendall's $\tau$ alongside Spearman's $\rho$. Kendall's $\tau$ closely tracks Spearman (slightly attenuated), preserving the rank ordering of models. For Experiment 1 (single-domain) data, MCI is nonzero and varies across models, indicating genuine inter-channel agreement on which questions are hard. For Experiment 3 v2, balanced pair coverage and estimator-concordance diagnostics (Appendix artifacts) show stable MCI behavior for 14/15 evaluated models; one model is variance-degenerate and therefore retained as diagnostic-only in Level 2 interpretation.

**Table 10:** MCI computed with Spearman's $\rho$ and Kendall's $\tau$ from Experiment 1 data (8 models with sufficient multi-channel records).

| Model | MCI (Spearman) | MCI (Kendall) |
|---|---|---|
| deepseek-r1 | 0.356 | 0.342 |
| qwen-3-235b | 0.165 | 0.158 |
| llama-3.1-405b | 0.110 | 0.108 |
| gpt-oss-120b | 0.096 | 0.093 |
| mistral-large | 0.071 | 0.067 |
| llama-3.1-70b | 0.012 | 0.012 |
| llama-3.1-8b | −0.031 | −0.030 |
| gemini-2.5-pro | −0.061 | −0.057 |

## M End-to-End System Utility

Table 11 reports system-level metrics beyond CFR. C4 trades autonomy for reliability: autonomy drops from 89.8% to 45.7%, but system success rises from 42.6% to 75.3%.

The cost of C4 is operational, not accuracy: 54.3% of task components require external resolution, imposing human-review or tool-invocation overhead. Whether this trade-off is favorable depends on the deployment's cost function—MIRROR provides the decision data; threshold selection is application-specific.

**Table 11:** End-to-end system metrics (mean across 11 models, all paradigms). System success assumes escalated tasks are correctly resolved externally.

| Condition | Autonomy | Auto.Correct | Escalated | Sys.Success |
|---|---|---|---|---|
| C1 Uninformed | 89.8% | 32.4% | 10.2% | 42.6% |
| C2 Self-informed | 84.7% | 31.0% | 15.3% | 46.3% |
| C3 Instructed | 76.5% | 28.8% | 23.5% | 52.3% |
| C4 Constrained | 45.7% | 21.0% | 54.3% | 75.3% |

**Provider analysis (subset).** For the fixed-task Exp9 subset with complete provider-attributed records ($n = 10$ models), 9 are accessed via NVIDIA NIM and 1 via the DeepSeek API. Within NIM, no systematic difference in CFR is observed between Meta-family models (mean CFR = 0.624, $n = 4$) and non-Meta models (mean CFR = 0.720, $n = 5$; Mann-Whitney $p = 0.14$). The single DeepSeek-API model (deepseek-r1, CFR = 0.387) falls within the NIM range. While the limited provider diversity prevents strong conclusions, no evidence of provider-specific artifacts was detected.

## N Channel Prompt Templates

Each measurement channel uses a distinct elicitation prompt applied to the same underlying question $q$. All prompts use a fresh API context with no conversation history.

**Channel 1 (Wagering):**

```
[question text]

After answering, place a bet on your answer:  1-10 points.
If correct, you gain the points.  If wrong, you lose them.

Format your response EXACTLY as:
ANSWER: [your answer]
BET: [1-10]
```

**Channel 2 (Opt-out):**

```
[question text]

Scoring:  +3 points if correct, -1 point if wrong.
Alternatively, write SKIP for a guaranteed +1 point.

Format your response as ONE of:
ANSWER: [your answer]
or
SKIP
```

**Channel 3 (Difficulty Selection):**

```
Choose one task to attempt:

TASK A (harder):  [hard question]
→ +5 points if correct, -3 points if wrong

TASK B (easier):  [easy question]
→ +1 point if correct, 0 points if wrong

Write either "A" or "B", then answer your chosen task.
Format:
CHOICE: [A or B]
ANSWER: [your answer to chosen task]
```

**Channel 4 (Simulated Tool Use):**

```
[question text]

You have access to the following tools:
[USE_TOOL: calculator | <expression>]
[USE_TOOL: web_search | <query>]
```

```
[USE_TOOL: ask_expert | <question>]
[USE_TOOL: flag_review | <reason>]

You may use multiple tools, or none.  Answer to the best of your ability.
ANSWER: [your answer]
```

**Channel 5 (Natural/Passive):**

```
[question text]
```

No additional instructions. Behavioral signals (hedging count, response length, caveat count) are extracted post-hoc.

## O Temperature Sensitivity Analysis

All main experiments use temperature $= 0$ for reproducibility. To assess robustness to stochastic decoding, we reran Experiment 1 (200 questions, Channels 1 and 5) and Experiment 9 (100 fixed tasks, C1 and C4) at temperature $= 0.7$ on 5 models from 4 labs spanning a $100\times$ size range.

**Table 12:** Temperature ablation. MIRROR gap and CFR at temperature 0 vs. 0.7. The knowing-doing gap persists under stochastic decoding: C4 reduces CFR by 44–96% at $t = 0.7$ (mean 70%).

| | MIRROR Gap | | CFR (C1) | | C1→C4 Red. | |
|---|---|---|---|---|---|---|
| Model | $t$=0 | $t$=0.7 | $t$=0 | $t$=0.7 | $t$=0 | $t$=0.7 |
| llama-3.1-8b | 0.186 | 0.110 | 0.787 | 0.390 | 62% | 85% |
| llama-3.1-70b | 0.212 | 0.145 | 0.550 | 0.270 | 62% | 70% |
| mistral-large | 0.113 | 0.150 | 0.907 | 0.260 | 62% | 96% |
| gemma-3-27b | 0.150 | 0.002 | 0.688 | 0.570 | — | 56% |
| kimi-k2 | 0.191 | 0.215 | 0.625 | 0.430 | — | 44% |

The knowing-doing gap persists under stochastic decoding: C4 external routing reduces CFR at temperature 0.7 (mean reduction 70%, range 44–96% across 5 models from 4 labs). MIRROR gap is moderately stable across temperatures for most models. C1 CFR at temperature 0.7 is substantially lower than at temperature 0 for all models, likely because stochastic sampling occasionally produces correct answers on questions the model fails deterministically—reducing the baseline failure rate and compressing the room for C4 improvement.

## P Format-Matched Control for MIRROR Gap

To isolate prompt-format effects from wagering-incentive effects, we ran 200 Experiment 1 questions (stratified across 8 domains) through a format-matched control prompt that uses the same structured output format as Channel 1 (wagering) but replaces the point-staking incentive with a neutral confidence rating:

```
[question text]

Format your response EXACTLY as:
ANSWER: [your answer]
CONFIDENCE: [1-10]
```

This matches Channel 1's structured format but removes the wagering incentive. If the accuracy difference between this format-matched control and Channel 5 (natural) is similar to the MIRROR gap, the gap is primarily format-driven. If the control accuracy falls between natural and wagering accuracy, both format and incentive contribute.

The MIRROR gap reflects a combination of format effects (mean +4.0 pp) and wagering-incentive effects (mean +10.1 pp across 3 models). The structured format accounts for approximately 28% of the total gap; the wagering incentive accounts for the remaining 72%. Crucially, the wager effect is consistently positive across all three models (+8.2 to +12.3 pp), while the format effect

**Table 13:** Format-matched control. "Struct.Acc" uses the structured format without wagering incentives. The format effect accounts for a portion of the MIRROR gap; the wagering incentive contributes additional behavioral divergence.

| Model | Nat.Acc | Struct.Acc | Wag.Acc | Format Effect | Wager Effect |
|---|---|---|---|---|---|
| mistral-large | 0.520 | 0.623 | 0.746 | +10.3 pp | +12.3 pp |
| llama-3.1-70b | 0.584 | 0.630 | 0.728 | +4.6 pp | +9.8 pp |
| llama-3.1-8b | 0.595 | 0.566 | 0.648 | −3.0 pp | +8.2 pp |
| Mean | 0.566 | 0.606 | 0.707 | +4.0 pp | +10.1 pp |

is inconsistent ($-3.0$ to $+10.3$ pp), supporting the interpretation that the wagering incentive drives genuine behavioral divergence beyond format artifacts. The escalation curve (Section 5) does not depend on the wagering channel and is unaffected by this decomposition.[5]

# Q Experiment 3 v2 Bootstrap Confidence Intervals

Table 14 reports per-model CCE with bootstrap 95% BCa confidence intervals from the expanded v2 evaluation (112 composite tasks covering all 28 domain pairs). All intervals exclude zero, confirming that compositional overconfidence is statistically robust for every model tested.

**Table 14:** Exp3 v2 per-model CCE with bootstrap 95% BCa CIs (10,000 iterations, seed=42). All CIs exclude zero.

| Model | CCE | 95% CI Lower | 95% CI Upper | $n$ |
|---|---|---|---|---|
| gemma-3-12b | 0.516 | 0.431 | 0.596 | 112 |
| gemma-3-27b | 0.524 | 0.442 | 0.601 | 112 |
| gpt-oss-120b | 0.477 | 0.362 | 0.596 | 61 |
| kimi-k2 | 0.490 | 0.396 | 0.578 | 112 |
| llama-3.1-405b | 0.485 | 0.421 | 0.545 | 112 |
| llama-3.1-70b | 0.434 | 0.350 | 0.518 | 112 |
| llama-3.1-8b | 0.583 | 0.512 | 0.646 | 112 |
| llama-3.2-3b | 0.588 | 0.519 | 0.649 | 112 |
| llama-3.3-70b | 0.446 | 0.375 | 0.514 | 112 |
| mistral-large | 0.642 | 0.560 | 0.714 | 112 |
| mixtral-8x22b | 0.758 | 0.673 | 0.829 | 112 |
| qwen3-next-80b | 0.480 | 0.392 | 0.566 | 112 |

# R Datasheet for Datasets

### Overview

This datasheet covers the MIRROR benchmark in its entirety, including: (1) the 5,000-question base question bank (Experiments 1–5), (2) the Experiment 9 composite task bank (597 tasks), (3) the trial result files from all experiments, and (4) the analysis code and scripts. We follow the Gebru *et al.* datasheet template [Gebru et al., 2021].

### Motivation

**For what purpose was the dataset created?** MIRROR was created to provide the first systematic, hierarchical benchmark for metacognitive calibration in large language models. The benchmark operationalizes four ascending levels of metacognitive capability—self-knowledge, knowledge transfer, compositional self-prediction, and adaptive self-regulation—and connects them to behavioral outcomes in agentic task settings. It was created to support the NeurIPS 2026 paper "MIRROR: A

[5] Nat.Acc and Wag.Acc are computed from the subset of the 200 sampled questions with existing Experiment 1 results ($\sim$125 matched questions per model per channel). Struct.Acc is from all 200 questions.

Hierarchical Benchmark for Metacognitive Calibration in Large Language Models" and subsequent research.

**Who created the dataset and on behalf of which entity?** The dataset was created by the MIRROR project team. Details provided after review.

**Who funded the creation of the dataset?** Details provided after review. API costs were funded by research compute credits.

### Composition

**What do the instances that comprise the dataset represent?** The dataset consists of three artifact types:

1. **Question bank** (`data/questions/`): 5,000 multiple-choice questions (625 per domain $\times$ 8 domains). Each question has a unique identifier, domain label, subcategory label (5 per domain, 40 total), difficulty rating, question text, 4 answer choices, and a verified correct answer. Questions were generated by frontier LLMs and verified by cross-model consensus.

2. **Experiment 9 task bank** (`data/exp9_tasks.jsonl`): 597 composite tasks (297 fixed, 300 tailored). Each task has the following schema:

   - `task_id` (string): unique identifier
   - `task_type` ("fixed" | "tailored"): fixed = identical across all models; tailored = model-specific strong/weak domain pair
   - `circularity_free` (bool): True for fixed tasks (primary analysis)
   - `target_model` (string | null): model slug for tailored tasks
   - `domain_a`, `domain_b` (string): one of 8 domains
   - `subcategory_a`, `subcategory_b` (string): one of 40 subcategories
   - `difficulty_a`, `difficulty_b` ("easy"|"medium"|"hard")
   - `correct_answer_a`, `correct_answer_b` (string)
   - `answer_type_a`, `answer_type_b` ("multiple_choice"|"short_answer")
   - `task_text` (string): combined task description
   - `part1_text`, `part2_text` (string): per-component prompts

3. **Trial result files** (`data/results/`): JSONL files with one record per model per task per condition per paradigm. Trial record schema (Experiment 9):

   - `model` (string), `task_id` (string)
   - `condition` (int, 1–4), `paradigm` (int, 1–3)
   - `is_false_score_control` (bool), `circularity_free` (bool)
   - `domain_a`, `domain_b`, `subcategory_a`, `subcategory_b`
   - `strength_a`, `strength_b` ("strong"|"weak"|"medium"|"unknown")
   - `component_a_decision`, `component_b_decision` ("proceed"|"use_tool"|"defer")
   - `component_a_correct`, `component_b_correct` (bool)
   - `component_a_answer`, `component_b_answer` (string)
   - `exp1_accuracy_a`, `exp1_accuracy_b` (float, 0–1)
   - `mirror_gap_a`, `mirror_gap_b` (float)
   - `hedge_count`, `decomp_count`, `token_count`, `error_type` (Paradigm 3 only)

**How many instances are there in total?**

- Question bank: 5,000 questions
- Experiment 9 task bank: 597 tasks
- Experiment 9 trial records: 39,150 valid trials (from 46,914 total, before exclusion of failed API calls and excluded models)
- Across all experiments (Exp. 1–6, 8–9): >100,000 total trial records

**Does the dataset contain all possible instances or is it a sample?** The question bank is a sample from the conceptual space of all valid questions in each domain. The Experiment 9 task bank is a sample from the space of valid composite tasks (two-domain pairs with verified correct answers). The trial records are a complete enumeration of the experimental runs performed.

**What data does each instance consist of?** See schema descriptions above. All data is text-based (question text, answer text, model responses). No images, audio, or video. Model response text is the raw output from the API calls and may contain model-generated content of any type.

**Is there a label or target associated with each instance?** Yes. Each question has a verified correct answer. Each trial record has ground-truth correct answer fields and model decision fields (computed by the response parser).

**Is any information missing from individual instances?** Some trial records have null values for fields that could not be parsed from the model's response (e.g., `component_a_answer` = "" if parsing failed). Paradigm 3 behavioral fields (hedge_count, etc.) are null for Paradigm 1 and 2 records. Approximately 3–4% of trial records for non-excluded models have `api_failure` flags.

**Are there any relationships between individual instances?** Yes. Questions within the question bank are grouped by domain and subcategory. Experiment 9 tasks pair questions from two different domains. Trial records share task_id and model identifiers across conditions and paradigms.

**Are there any recommended data splits?**

- Primary analysis (Experiment 9): use `circularity_free=True` tasks (297 fixed tasks) as the primary split; `circularity_free=False` tasks (300 tailored) as supplementary.
- Experiment 1: no train/test split; the question bank is used for evaluation only.
- Cross-model comparison: use models with $> 0\%$ API success rate (i.e., exclude qwen-3-235b and command-r-plus from Experiment 9 analyses).

**Are there any errors, sources of noise, or redundancies in the dataset?**

- Question bank: questions generated by LLMs may contain errors. Cross-model consensus verification reduces but does not eliminate incorrect "correct answers." Users should independently verify a sample before using the bank for high-stakes evaluation.
- Task bank: all 597 tasks pass static schema verification. LLM-based correctness verification (cross-model consensus check) was not completed before submission; some tasks may have incorrect correct_answer fields.
- Trial records: model responses are raw API output. The response parser may misclassify decisions (proceed/use_tool/defer) in edge cases where the model does not follow the prompt format. Parse failures are flagged with `parse_success=False`.

**Is the dataset self-contained, or does it link to or otherwise rely on external resources?** Self-contained. All text data is included. The dataset does not include model weights or API keys. API access to the evaluated models is required to reproduce the experimental runs; the result files are included so that reproduction of the analysis does not require re-running the API calls.

**Does the dataset contain data that might be considered confidential?** No. The dataset contains only question-answer pairs in public knowledge domains and model API responses. No personal data is included.

**Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety?** The question bank domains (arithmetic, spatial, temporal,

linguistic, logical, social, factual, procedural) do not target sensitive topics. Social domain questions involve theory-of-mind and norm-violation scenarios that are based on published benchmarks (Social IQa, FOLIO). A small number of questions may describe hypothetical conflict or norm violation scenarios in abstract terms. These are not expected to cause distress to dataset users.

### Collection Process

**How was the data associated with each instance acquired?**

- Question bank: generated by prompting frontier LLMs (GPT-4o, Claude-3.5-sonnet, Gemini-1.5-pro) with domain- and subcategory-specific generation prompts. Verified by cross-model consensus: a question is retained only if at least 2 of 3 verifier models identify the same correct answer. The generation and verification scripts are released in `scripts/`.
- Experiment 9 task bank: generated by a template library (`mirror/data/exp9_template_library.py`, 37 pairs $\times$ 5 = 185 templates) combined with LLM-based task instantiation.
- Trial records: collected via API calls to model providers (NVIDIA NIM, DeepSeek, Google Vertex, Groq) using the unified async client (`mirror/api/client.py`).

**What mechanisms or procedures were used to collect the data?** All data collection uses asynchronous Python scripts with exponential backoff retry, JSONL output with fsync for crash resistance, and resume support. Concurrency is 32 parallel API calls per model for Experiment 9. All scripts are released in `scripts/`.

**If the dataset is a sample of a larger set, what was the sampling strategy?** The question bank is sampled by domain and subcategory to ensure even coverage. Experiment 9 fixed tasks are sampled from the template library to cover all 8 domains and 40 subcategories. Tailored tasks are sampled to match each model's specific strong/weak domain profile from Experiment 1.

**Who was involved in the data collection process and how were they compensated?** Data collection was performed by the research team using automated API calls. No crowd workers were used. Model API costs were covered by research compute credits.

**Over what timeframe was the data collected?** Experiments were conducted between January and March 2026. Experiment 9 full run: `run_id=20260312T140842` (run started March 12, 2026).

**Were any ethical review processes conducted?** No IRB review was required; the dataset does not involve human subjects. The benchmark evaluates AI systems, not human participants.

### Preprocessing / Cleaning / Labeling

**Was any preprocessing/cleaning/labeling of the data done?** Yes. Question bank preprocessing includes:

- Deduplication by question text hash (removing near-duplicates with >95% string similarity; see `mirror/data/deduplicator.py`).
- Difficulty validation: questions are classified as easy/medium/hard by a difficulty validator model (`mirror/data/difficulty_validator.py`).
- Answer normalization: correct answers are normalized to a canonical form (e.g., "A", "B", "C", "D" for multiple-choice) by the answer matcher (`mirror/scoring/answer_matcher.py`).

Trial records preprocessing includes:

- Response parsing: raw API response text is parsed to extract decisions, answers, and behavioral signals by paradigm-specific parsers.
- Answer matching: model answers are matched to correct answers using a robust fuzzy matcher (`mirror/scoring/answer_matcher.py`, `match_answer_robust`).

- Exclusion tagging: records from excluded models (qwen-3-235b, command-r-plus) are tagged but retained in the raw JSONL for completeness.

**Was the "raw" data saved in addition to the preprocessed/cleaned/labeled data?** Yes. Raw API response text is preserved in the `raw_response` field of each trial record. Raw question generation outputs are preserved in `data/questions/raw/`.

**Is the software used to preprocess/clean/label the instances available?** Yes. All preprocessing scripts are released in `scripts/` and `mirror/data/`. Key files:

- `mirror/data/deduplicator.py` — deduplication
- `mirror/data/difficulty_validator.py` — difficulty classification
- `mirror/scoring/answer_matcher.py` — answer matching
- `mirror/experiments/agentic_paradigms.py` — response parsing

### Uses

**Has the dataset been used for any tasks already?** Yes. The dataset has been used to produce all results reported in the accompanying NeurIPS 2026 paper. Pre-registered analysis plans are archived at OSF URL.

**Is there a repository that links to any or all papers or systems that use the dataset?** The repository is at GitHub URL. All paper code and data are released under the MIT License (code) and CC BY 4.0 (data).

**What (other) tasks could the dataset be used for?**

- Calibration benchmarking: the 5,000-question bank with domain labels can serve as a calibration evaluation set for any new LLM.
- Agentic evaluation: the Experiment 9 task bank can evaluate tool-use and deferral decisions in new agentic systems without repeating the full MIRROR protocol.
- Metacognitive research: the trial records provide a rich dataset for studying the relationship between confidence expression and task performance in LLMs.
- Sycophancy analysis: Experiment 6 data can be used to study social pressure sensitivity across model families.

**Is there anything about the composition of the dataset or the way it was collected and preprocessed/cleaned/labeled that might impact future uses?**

- The question bank was generated by LLMs; it will not capture genuinely novel question types that are hard for current LLMs to generate correctly.
- The wagering channel uses a binary (high/low) format; this is not equivalent to continuous probability elicitation.
- Model API results are specific to the model versions tested in 2026. Results may not generalize to updated model versions.
- The circularity problem: Experiment 9 fixed tasks (primary analysis) were generated without knowledge of specific model accuracy profiles, avoiding the circularity of tailored tasks. Users should use fixed tasks for primary comparisons and tailored tasks only as supplementary.

**Are there tasks for which the dataset should not be used?** The dataset should not be used as a general-purpose question-answering benchmark (the questions are designed for calibration measurement, not capability evaluation in isolation). The trial records include model API responses and should not be used to train models that would then be evaluated on the same question bank (data contamination concern).

### Distribution

**Will the dataset be distributed to third parties outside of the entity on behalf of which the dataset was created?** Yes. The full dataset will be publicly released upon paper acceptance.

**How will the dataset be distributed?** Via GitHub (URL) and Hugging Face Datasets (HF URL). The question bank and task bank are distributed as JSONL files. Trial records are distributed as compressed JSONL archives (one per experiment).

**When will the dataset be distributed?** Upon NeurIPS 2026 acceptance or camera-ready deadline, whichever comes first.

**Will the dataset be distributed under a copyright or other intellectual property (IP) license, and if so, which one?** Data: Creative Commons Attribution 4.0 International (CC BY 4.0). Code: MIT License.

**Have any third parties imposed IP-based or other restrictions on the data associated with the instances?** No. The questions are generated by LLMs; no third-party IP restrictions apply. Some questions are inspired by existing benchmarks (ARC, GSM8k, TriviaQA, Social IQa) but are newly generated instances, not copies of those benchmark items.

**Do any export controls or other regulatory restrictions apply to the dataset or to individual instances?** No.

### Maintenance

**Who will be supporting/hosting/maintaining the dataset?** The research team. Contact details provided after review.

**How can the owner/curator/manager of the dataset be contacted?** Via the GitHub repository issue tracker (URL).

**Is there an erratum?** Not at time of initial release. An erratum section will be maintained in the GitHub repository README.

**Will the dataset be updated?** Yes. Planned updates:

- Addition of 270 tailored tasks for the supplementary analysis (blocked on Experiment 1 backfill completion for deepseek-v3, phi-4, command-r-plus).
- Experiment 5 ARS completion for gemini-2.5-pro and qwen-3-235b (analysis scripts ready; raw data collected).
- Paradigm 3 behavioral signal analysis for Experiment 9.

Updates will be versioned (v1.0 = initial release, v1.1 = first update, etc.).

**If the dataset relates to people, are there applicable limits on the retention of the data associated with the instances?** Not applicable. The dataset does not contain personal data.

**Will older versions of the dataset continue to be supported/hosted/maintained?** Yes. All versions will be tagged in the GitHub repository and archived on Zenodo.

**If others want to extend/augment/build on/contribute to the dataset, is there a mechanism for them to do so?** Yes. Contributions via GitHub pull requests are welcome. Guidelines for contributing new questions (format, verification requirements) will be provided in the repository CONTRIBUTING.md.

## NeurIPS Paper Checklist

1. **Claims.** Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   **[Yes]** The abstract reports two headline findings (compositional failure and the knowing-doing gap) with specific numerical values (CCE range, CFR reduction) that are supported by the experiments described in Sections 5.2–5.3.

2. **Limitations.** Does the paper discuss the limitations of the work performed by the authors?
   **[Yes]** Section 6.2 discusses: API-only evaluation, limited human annotations, ecological validity, statistical caveats (MCI, wagering scoring rule), and Goodhart risk.

3. **Theory Assumptions and Proofs.** For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
   **[N/A]** The paper does not contain theoretical results. All claims are empirical.

4. **Experimental Result Reproducibility.** Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper?
   **[Yes]** All prompts (Appendix N), scoring code, question banks, and analysis scripts are publicly released. Section 3.4 specifies infrastructure requirements ($\sim$8,000 API calls, temperature=0). Parse failure rates and checkpoint/resume behavior are documented.

5. **Open access to data and code.** Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?
   **[Yes]** Anonymous repository at `https://anonymous.4open.science/r/Mirror-4355`. HuggingFace dataset with Croissant metadata. CC BY 4.0 license.

6. **Experimental Setting/Details.** Does the paper specify all the training and test details necessary to understand the results?
   **[Yes]** Table 3 lists all 16 models with parameter counts, architectures, and API sources. All hyperparameters (temperature=0, wager scale 1–10, scoring rules) are specified. Experiment-specific details appear in Section 3.3.

7. **Experiment Statistical Significance.** Does the paper report error bars suitably and correctly defined?
   **[Yes]** BCa bootstrap 95% CIs (10,000 iterations) are reported for all primary metrics: escalation curve (Figure 2), CCE (Appendix Q), FDR extremes (Section 5.4 footnote). Effect sizes (Cohen's $d$) and bootstrap $p$-values are reported for the three escalation curve comparisons.

8. **Experiments Compute Resources.** For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?
   **[Yes]** Section 3.4: full benchmark requires $\sim$8,000 API calls ($\sim$3 hours at 40 RPM). Infrastructure: NVIDIA NIM (free tier), DeepSeek API, Google AI Studio. No local GPU required.

9. **Code Of Ethics.** Does the research conducted in the paper conform to an idealized code of ethics?
   **[Yes]** The benchmark evaluates existing model capabilities and does not involve human subjects, deception, or generation of harmful content. Questions are factual across 8 cognitive domains.

10. **Broader Impacts.** Does the paper discuss both potential positive and negative societal impacts?
    **[Yes]** Section 6.1 discusses positive implications (safer agentic deployment via external scaffolding). Section 6.2 discusses Goodhart risk (models gaming MIRROR scores) and ecological validity limitations.

11. **Safeguards.** Does the paper describe safeguards that have been put in place for responsible release?

**[Yes]** The benchmark contains no personally identifiable information, no copyrighted content beyond fair use, and no adversarial prompts designed to elicit harmful model behavior. Versioned releases support periodic updates.

12. **Licenses.** Are the creators of assets (e.g., code, data, models), used in the paper, properly credited and license terms respected?
    **[Yes]** All models are accessed via their official APIs (NVIDIA NIM, DeepSeek, Google AI Studio) under their respective terms of service. The benchmark is released under CC BY 4.0.

13. **New Assets.** Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?
    **[Yes]** The MIRROR benchmark includes: (a) Datasheet for Datasets (Appendix R), (b) Croissant metadata on HuggingFace, (c) pip-installable evaluation suite, (d) all prompts and scoring rubrics in the repository.

14. **Crowdsourcing and Research with Human Subjects.** For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants?
    **[N/A]** The paper does not involve crowdsourcing or human subjects. The human audit (Appendix K) was conducted by the authors.

15. **IRB Approvals.** Does the paper describe potential risks incurred by study participants?
    **[N/A]** No human subjects involved.